NOT MEASUREMENT SENSITIVE

O-M-232P
26 September 2024
SUPERSEDING
O-M-232N
8 February 2016

FEDERAL SPECIFICATION

METHANOL (METHYL ALCOHOL)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three grades and four unit quantities of methanol.

1.2 Classification. Methanol shall be of the following grades and unit quantities (see 6.2):

1.2.1 Grade.

Grade A = grade A - Synthetic, 99.85 percent by weight (solvent use)
Grade B = grade AA - Synthetic, 99.85 percent by weight (hydrogen-carbon dioxide generation use)
Grade C = grade C - Wood alcohol (denaturing grade)

1.2.2 Unit quantity.

1-QT - 1 quart
1-GL - 1 gallon
5-GL - 5 gallons
54-GL - 54 gallons

1.3 International standardization agreement code number. O-M-232, Grade A methanol is identified by NATO Code S-747.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 6090 Strathmore Rd, Richmond, VA 23237 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Code of Federal Regulations (CFR):

27 CFR 21	-	Formulas for Denatured Alcohol and Rum
29 CFR 1910.1200	-	Occupational Safety and Health Standards, Hazard Communication

(Electronic copies of the CFR may be obtained from https://www.ecfr.gov.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ASTM International Standards:

ASTM D1078	-	Standard Test Method for Distillation Range of Volatile Organic Liquids
ASTM D1209	-	Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)
ASTM D1296	-	Standard Test Method for Odor of Volatile Solvents and Diluents
ASTM D1353	-	Test Method for Nonvolatile Matter in Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products
ASTM D1363	-	Standard Test Method for Permanganate Time of Acetone and Methanol
ASTM D1613	-	Standard Test Method for Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products
ASTM D1722	-	Standard Test Method for Water Miscibility of Water-Soluble Solvents
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4629	-	Standard Test Method for Trace Nitrogen in Liquid Hydrocarbons by Syringe/Inlet Oxidative Combustion and Chemiluminescence Detection
ASTM D5386	-	Standard Test Method for Color of Liquids Using Tristimulus Colorimetry
ASTM D6304	-	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fisher Titration
ASTM D7184	-	Standard Test Method for Ultra Low Nitrogen in Aromatic Hydrocarbons by Oxidative Combustion and Reduced Pressure Chemiluminescence Detection

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ASTM D8005	-	Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)
ASTM E203	-	Standard Test Method for Water Using Volumetric Karl Fischer Titration
ASTM E346	-	Standard Test Method for Analysis of Carbonizables in Methanol
ASTM E1064	-	Standard Test Method for Water in Organic Liquids by Coulometric Karl Fischer Titration
ASTM E2680	-	Standard Test Method for Appearance of Clear, Transparent Liquids (Visual Inspection Procedure)

(Electronic copies of ASTM documents may be obtained from https://www.astm.org.)

International Methanol Producers and Consumers Association (IMPCA) Specifications:

IMPCA 001	-	IMPCA methanol reference specification

(Electronic copies of IMPCA documents may be obtained from https://www.impca.eu.)

3. REQUIREMENTS

3.1 Grades A and AA. Grades A and AA methanol shall conform to the chemical and physical characteristics of table I when tested as specified therein.

3.2 Grade C. Grade C methanol shall comply with the requirements for methyl alcohol used as a denaturant in 27 CFR 21.

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TABLE I. Chemical and physical characteristics of grades A and AA methanol.

Characteristic	Requirements		Test method[1]
	Grade A	Grade AA	
Acetone, percent by mass, maximum	0.003	0.002	IMPCA 0001
Acidity (as acetic acid), percent by mass, maximum	0.003	0.003	ASTM D1613
Appearance	Bright and clear	Bright and clear	ASTM E2680
Carbonizable impurities, color, Platinum-Cobalt (Pt-Co), maximum	No. 30	No. 30	ASTM E346
Color, Pt-Co, maximum	No. 5	No. 5	ASTM D1209, D5386, or D8005
Distillation range at 760 mm Hg, maximum	1.0 °C (and shall include 64.6 ± 0.1 °C)	1.0 °C (and shall include 64.6 ± 0.1 °C)	ASTM D1078
Ethanol, percent by mass, maximum	--	0.001	IMPCA 001
Nonvolatile matter, mg per 100 mL, maximum	10	10	ASTM D1353
Odor	Characteristic, non-residual	Characteristic, non-residual	ASTM D1296
Permanganate time at 15 °C	No discharge of color in 30 minutes	No discharge of color in 30 minutes	ASTM D1363
Specific gravity at 20/20 °C, maximum	0.7928	0.7928	ASTM D4052
Total nitrogen, mg/kg maximum[2]	1.0	1.0	D4629 or D7184
Water, percent by mass, maximum	0.15	0.10	ASTM E1064, D6304, or E203
Water Miscibility	Pass	Pass	ASTM D1722

[1]In cases of dispute, the first method listed shall be the referee method.
[2]Testing for total nitrogen is not required but may be used in lieu of odor testing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the acquisition, the contractor is responsible for the performance of all inspection requirements as specified herein. Unless otherwise specified in the acquisition order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The methanol shall meet the requirements of table I.

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5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the acquisition order (see 6.2).

5.2 Precautionary marking. Each unit and shipping container shall be marked or labeled, as applicable, in accordance with 29 CFR 1910.1200(f) to show the required precautionary information. Each unit shall be marked to show the skull and crossbones poison symbol and the word "POISON." Each outer container shall be marked to show the top of the container by use of and arrow and the word "UP."

6. NOTES

INFORMATION FOR GUIDANCE ONLY. This section contains information of a general or explanatory nature that is helpful, but is not mandatory.

6.1 Intended use. Grade A methanol is intended for use as an industrial solvent. Grade AA methanol is intended for use in the generation of hydrogen and carbon dioxide. Grade C methanol (wood alcohol) is a product of the destructive distillation of wood and is intended for use as a denaturant.

6.2 Acquisition requirements. Acquisition documents must specify the following:

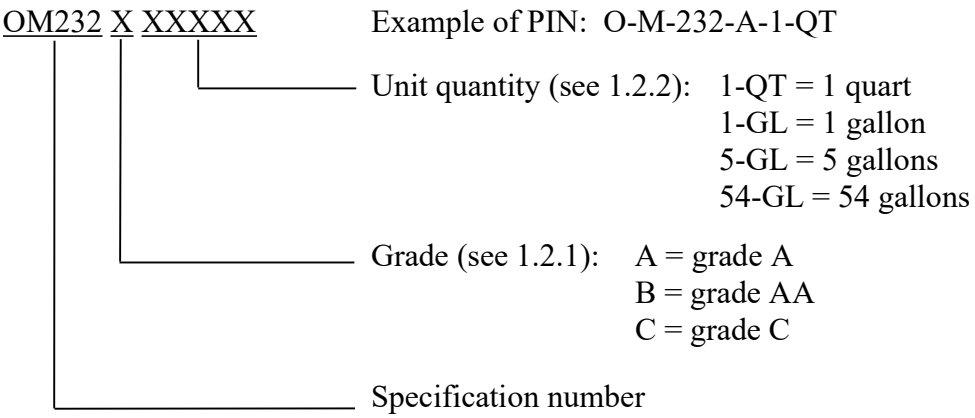
a. Title, number, and date of this specification.
b. Grade of methanol required (see 1.2.1).
c. Unit quantity required (see 1.2.2).
d. Responsibility for inspection, if different (see 4.1).
e. Inspection facilities, if different (see 4.1).
f. Packaging (see 5).

6.3 Safety data sheet (SDS). Contracting officers will identify those activities requiring copies of completed SDS prepared in accordance with FED-STD-313 and meeting the requirements of 29 CFR 1910.1200. The pertinent government mailing addresses for submission of the data are listed in FED-STD-313 and 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the SDS.

6.4 Submission of alternate inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

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6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27 Volumes 1 and 2. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.7 Subject term (key word) listing.

denaturant
industrial solvent
wood alcohol

6.8 International standardization agreements. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations. Identified below are the specific paragraph numbers and the international standardization agreements applicable to this specification:

AIR-STD-4024	-	Interchangeability Chart of Standardized Aviation Fuels, Lubricants, and Associated Products.
STANAG-1135	-	Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations.

6.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this

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document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:	CIVIL AGENCY COORDINATING ACTIVITY:
Custodians: Army - EA Navy - OS Air Force – 68 DLA – GS	GSA - FAS
Review Activities: Army - AR, GL, MD1, MI Navy - AS	Preparing Activity: DLA - GS3 (Project 6810-2024-014)

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